



Hong Kong University of Science & Technology

Department of Electronic & Computer Engineering

Spring 2026

ELEC 5160 Design Project

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(Due: May 19 by 18:00)

Submit your project online through Canvas before the deadline.

Objective

This design project is based on group work with 2 students permitted per group. You may also attempt the project individually, for which you will be given extra credit only if you successfully complete the implementation. You will be designing a full chip from specification phase down to layout. Several past publications on different digital architectures are shared for your reference. You may pick one of these papers for your project implementation. Alternatively, you may propose a design of your choice, but it will need to be approved by the course instructor first. Most of your project work needs to follow the RTL design flow approach. However, a very small portion may be based on full custom flow also. In this case, at the end, you will then need to stitch together the digital (full custom), digital (semi-custom), and any analog blocks in the Cadence analog environment to demonstrate full functionality.

Details

You are responsible for the Verilog code (plus full custom schematics, if any) to describe the hardware. You are also expected to write all related test-benches to ensure functional correctness and code coverage. Following the tutorial conducted by the TA, take your entire project through the design, pre-sim, synthesis, post-synth sim, automatic place and route (plus custom layout if any), and finally the post-layout sim stage.

Your grade will be based on the correctness, silicon footprint, performance, and power consumption of your implementation. In a group, the work must be distributed equally between the two members. You are responsible for the time management and workload distribution of the project. You will be required to sign a declaration indicating the relative amount of work (in %) each member of the group has contributed. For those attempting this project on their own (i.e. without a group member), you will be eligible for extra credit only if the project is completed successfully.

Technology

You are expected to implement your project using the commercial TSMC 180nm CMOS process, for which you have all signed an NDA already.

Report & Demonstration (25% of course grade)

Project demonstrations will take place on May 20 tentatively. Project reports (5%) are to be submitted via canvas by May 19. Timeslots for demonstration (roughly 20-30 minutes per group) and details of the venue will be posted on canvas.

The project demonstration (20%) will consist of a run-time execution of your design, and a Q&A session about your HDL code as well as your synthesis and layout results.

The demonstration will be focused on the following key points:

- Coding style and code quality including test-benches (4%)
- Response to questions (4%)
- Difficulty level (3%)
- Behavioral simulation (1%)
- Synthesis (2%)
- Post-synthesis simulation (2%)
- Layout (2%)
- Post-layout simulation (2%)
- Extra effort and/or special mention (bonus)

In your report, please include the following:

- i) Verilog Complete design and testbench HDL codes.
- ii) Waveforms of key signals from the pre-sim/behavioral-level stage.
- iii) Key reports and results from the synthesis step.
- iv) Waveforms of key signals from the post-synthesis-sim stage.
- v) A screenshot/image of your synthesized schematic.
- vi) A screenshot/image of your final placed and routed design.
- vii) Waveforms of key signals from the post-layout-sim stage.
- viii) The full path to your design and implementation files.